Homework 3 — PSet: 3. {10, 11, 13, 21}

Due 10.15.26 — Aaron Weymouth

10. Planet mass M; $e = 1 - \alpha$; $\alpha \ll 1$

- assume only gravity; if planet is struck at greatest distance from comet of mass m; in tangent dir.
  W/ inelastic collision, find KE_{min} to change to a parabolic orbit.

↳ Vis-vesa eq + conservation of Linear momentum + parabolic ESC.

Let $\mu = GM_\odot$

↙ a is planets OG semi-major ax.'s.

↳ aphelion: $r_a = a(1+e) = a(2-\alpha)$

$$v_a^2 = \mu\left(\frac{2}{r_a} - \frac{1}{a}\right) = \frac{\mu}{r_a}\left(2-(2-\alpha)\right)$$

↳ $$\boxed{v_a = \sqrt{\frac{\mu\alpha}{r_a}}} \quad \ldots (1)$$

Specific orbital energy: $\epsilon_f = \frac{1}{2}v_f^2 - \frac{\mu}{r_a}$

for parabola: $\epsilon_f \equiv 0$

$$\therefore \boxed{v_f = \sqrt{\frac{2\mu}{r_a}}} \quad \ldots (2)$$

Completely inelastic collision (comet + planet move together)

↳ tangental momentum conserved in impact.

↳ Min comet speed happens at $\sin(\pi/2) = 1 \rightarrow$ same dir.

$$Mv_a + mv_c = (M+m)v_f$$

$$v_c = \frac{(M+m)v_f - Mv_c}{m}$$ ↙ Sub in (1) + (2)

$$v_c = \frac{1}{m}\sqrt{\frac{\mu}{r_a}}\left((M+m)\sqrt{2} - M\sqrt{\alpha}\right)$$ Since $m \ll M$

$$\boxed{v_c \simeq \frac{M}{m}\sqrt{\frac{\mu}{r_a}}\left(\sqrt{2} - \sqrt{\alpha}\right)} \quad \ldots (3)$$

ch 3: p 10 cont.

Minimum KE $\quad KE_{min} = \frac{1}{2} m V_c^2$

$$KE_{min} \approx \frac{\mu M^2}{2 m r_a} \left(\sqrt{2} - \sqrt{\alpha}\right)^2 \quad \text{and} \quad r_a = a(2 - \alpha)$$

$$KE_{min} \approx \frac{G M_\odot M^2}{2 m a (2-\alpha)} \left(\sqrt{2} - \sqrt{\alpha}\right)^2$$

and for $\alpha \to 0$:

$$\boxed{KE_{min} \approx \frac{G M_\odot M^2}{2 m a}}$$

Sanity check: required comet energy scales as M^2/m checks out; small comet → large engy to change orbit.

↳ Parabolic orbit is $E = 0$ bndry between bound & unbnd regime; we only need enough tangential momentum to reach that bndry.

COM reduction + COE + Circular Kepler.

↳ 2 masses m_1, m_2 initially separated r_0 w period τ.

Both are stopped and released we want to Prove

$$t_{col} = \frac{\tau}{4\sqrt{2}}$$

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relative coordinate: $\vec{r} = \vec{r}_1 - \vec{r}_2$ and $\mu = G(m_1 + m_2)$

↳ relative EOM: $\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r}$

↳ before stopping the relative orbit is circular.

$$r_0 \omega^2 = \frac{\mu}{r_0^2}$$

thus:

$$\omega^2 = \frac{\mu}{r_0^3} \rightarrow \boxed{\tau = 2\pi\sqrt{\frac{r_0^3}{\mu}}} \quad \ldots (1)$$

After Particles stop: $\dot{r}(0) = 0$, $L = 0$; only radial left.

↳ COE per unit of mass (reduced mass)

$$\frac{1}{2}\dot{r}^2 - \frac{\mu}{r} = -\frac{\mu}{r_0}$$

↳ $\frac{1}{2}\dot{r}^2 = \mu\left(\frac{1}{r} - \frac{1}{r_0}\right)$ and bc particles fall inward

$$\boxed{\dot{r} = -\sqrt{2\mu\left(\frac{1}{r} - \frac{1}{r_0}\right)}} \quad \ldots (2)$$

Ch 3: P. 12 cont:

- fall time → integrate: $t_c = \int_0^{r_0} \frac{dr}{\sqrt{2\mu\left(\frac{1}{r} - \frac{1}{r_0}\right)}}$

Let $x = r/r_0$

↳ $t_c = \underbrace{\sqrt{\frac{r_0^3}{2\mu}}}_{1} \cdot \int_0^1 \underbrace{\sqrt{\frac{x}{(1-x)}}}_{2} dx$

Let $x = \sin^2\theta \rightarrow dx = 2\sin\theta\cos\theta\, d\theta$

2: $\int_0^1 \sqrt{\frac{x}{(1-x)}}\, dx = 2\int_0^{\pi/2} \sin^2\theta\, d\theta = \frac{\pi}{2}$

$\therefore \boxed{t_c = \frac{\pi}{2\cdot\sqrt{2}} \cdot \sqrt{\frac{r_0^3}{\mu}}}$ ...(3) now sub into ...(1) for orginal circular pd.

$$\boxed{t_c = \frac{\tau}{4\sqrt{2}}}$$

↳ Collision takes about $0.117\,\tau$ << one circular orbit period.

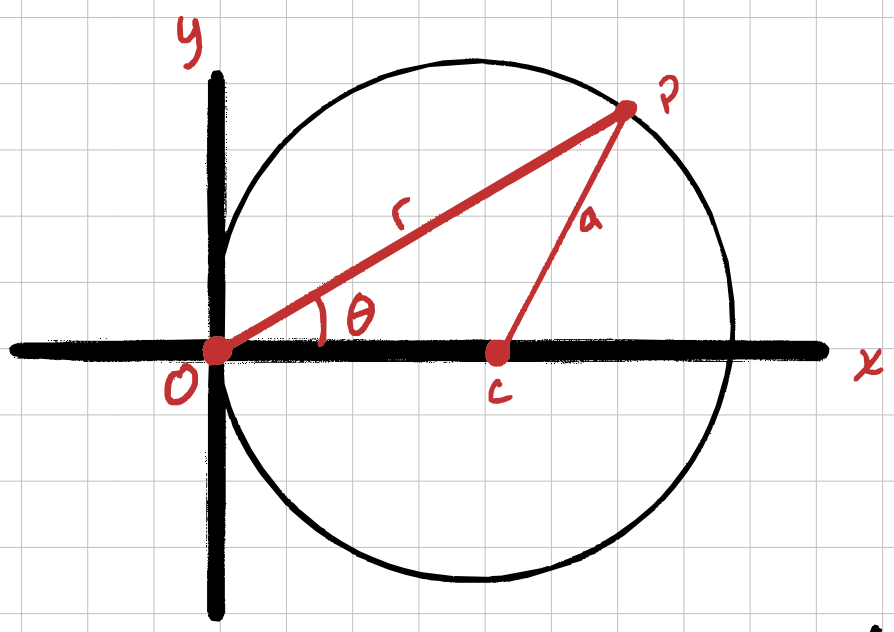
· once both particles are stopped → angular momentum vanishes

So 2 body problem becomes radial fall in 1D

Kepler's 3rd should indazzundantly arrive at result $r_0/2$

Ch 3: P. 13: Polar geo + con. angular momentum + Binet + energy

- Let attractive force be center $O = (0,0)$; circular orbit of rad a. centered at $C = (a, 0)$



$$(x-a)^2 + y^2 = a^2$$

$$x = r\cos\theta \; ; \; y = r\sin\theta$$

$$\hookrightarrow r^2 - 2ar\cos\theta = 0$$

for $r \neq 0$; $\boxed{r = 2 \cdot a \cdot \cos\theta}$ ...(1)

a. Show $F \propto r^{-5}$: Central force implies $\vec{L}$ conserved.

$h = r^2\dot{\theta}$ = same constant

and $u = 1/r \rightarrow F(r) = -mh^2u^2(u'' + u)$ (ch 3; sec 4)

$\hookrightarrow ' \rightarrow d/d\theta$

$u = 1/r = \frac{1}{2a}\sec\theta$

$\hookrightarrow u' = \frac{1}{2a} \cdot \sec\theta\tan\theta$ and $u'' = \frac{1}{2a}(\sec\theta\tan^2\theta + \sec^3\theta)$

$\swarrow 1 + \tan^2\theta$

$\therefore u'' + u = \frac{1}{2a} \cdot \sec\theta(\tan^2\theta + \sec^2\theta + 1) = \frac{1}{a} \cdot \sec^3\theta$

and $\sec\theta = 2 \cdot au$

$\hookrightarrow u'' + u = 8a^2u^3$

$\hookrightarrow F = -mh^2u^2(8a^2u^3) \rightarrow \boxed{F(r) = -\frac{8ma^2h^2}{r^5} = -\frac{k}{r^5}}$

* if $k = 8ma^2h^3 > 0$

Ch 3 P13 cont: b: Show total energy = 0

force: $F = -\frac{dU}{dr} = -\frac{k}{r^5}$; $U(\infty) = 0 \rightarrow \boxed{U(r) = -\frac{k}{4r^4}}$ ...(2)

KE: $T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta})$: since $\dot{r} = \frac{dr}{d\theta}\dot{\theta}$, $\dot{\theta} = \frac{h}{r^2}$

$$T = \frac{mh^2}{2r^4}\left[\left(\frac{dr}{d\theta}\right)^2 + r^2\right] \xrightarrow{r = 2a\cos\theta} \frac{dr}{d\theta} = -2a\sin\theta$$

then $\left(\frac{dr}{d\theta}\right)^2 + r^2 = 4a^2 \cdot (\sin^2\theta + \cos^2\theta) = 4a^2$

b $\boxed{T = \frac{2ma^2h^2}{r^4} = \frac{k}{4r^4}}$

Since $E = T + V = \frac{k}{4r^4} - \frac{k}{4r^4} \boxed{= 0 = E}$ Zero at every point: not just avg.

C: Period: Constant angular momentum → Constant areal velo:

$\frac{dA}{dt} = \frac{h}{2}$; full orbit $A = \pi a^2$ enclosed

$$\therefore T = \frac{A}{dA/dt} = \frac{\pi a^2}{h/2} = \boxed{\frac{2\pi a^2}{h} = 4\pi a^3 \cdot \sqrt{\frac{2m}{k}} = T}$$

No need to solve time dependance

P13 cont. δ: find $\dot{x}, \dot{y}, \dot{z}$ → shaw div

$$x = r\cos\theta = 2a\cos^2\theta$$
$$y = r\sin\theta = 2a\sin\theta\cos\theta$$

$$\dot{\theta} = \frac{h}{4a^2\cdot\cos^2\theta}$$

$$\Downarrow$$

$$\dot{x} = -4a\sin\theta\cos\theta\cdot\dot{\theta}$$

$$\dot{x} = -\frac{h}{a}\tan\theta$$

$$\dot{y} = 2a(\cos^2\theta - \sin^2\theta)\dot{\theta}$$

$$\dot{y} = \frac{h}{2a}(1-\tan^2\theta)$$

$$v = \sqrt{\dot{x}^2+\dot{y}^2} \rightarrow v^2 = \frac{h^2}{4a}\cdot(4\tan^2\theta + (1-\tan^2\theta)^2)$$

$$= (1+\tan^2\theta)^2 = \sec^4\theta \qquad \text{and if } h>0$$

$$\boxed{v = \frac{h}{2a}\sec^2\theta = \frac{2ah}{r^2}}$$

force center $r\to 0 \Leftrightarrow \theta \to \pm \pi/2$

$|\dot{x}|\to\infty$, $|\dot{y}|\to\infty$, $v\to\infty$ as particle approaches COF

P21: Effective pot + Binet + precession + Kepler 3rd

given: $V(r) = -\frac{k}{r} + \frac{h}{r^2}$; $k>0$; h is small perturb.

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① $V = V(r)$; $\ell = mr^2\dot{\phi} = \text{const}$

$$E = \frac{1}{2}m\dot{r}^2 + \frac{\ell^2}{2mr^2} - \frac{k}{r} + \frac{h}{r^2}$$

$$= \frac{1}{2}m\dot{r}^2 + \frac{\ell^2 + 2mh}{2mr^2} - \frac{k}{r}$$

$$\boxed{L^2 = \ell^2 + 2mh} \quad \ldots (1)$$

$$\boxed{E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} - \frac{k}{r}} \quad (2)$$

... (2) has same radial form as Kepler, but w/ effective angular momentum

radial → Keplerian :

② $F(r) = -\frac{dV}{dr} = -\frac{k}{r^2} + \frac{2h}{r^3}$ $\xrightarrow{\text{Binet}}$ $u'' + u = -\frac{m}{\ell^2 u^2}\cdot F\left(\frac{1}{u}\right)$, $u = 1/r$

$$u'' + u = \frac{mk}{\ell^2} - \frac{2mh}{\ell^2}u \rightarrow \boxed{u'' + \left(1 + \frac{2mh}{\ell^2}\right)u = \frac{mk}{\ell^2}} \quad \ldots (3)$$

Let

$B^2 = 1 + \frac{2mh}{\ell^2}$ $\rightarrow u'' + B^2 u = \frac{mk}{\ell^2}$ ★ I looked up diffeq sol.

$$u(\phi) = \frac{mk}{\ell^2 B^2}\left[1 + e\cos B(\phi - \phi_0)\right]$$

$$\hookrightarrow \boxed{r(\phi) = \frac{L^2/(mk)}{1 + e\cdot\cos(B(\phi - \phi_0))}} \quad \ldots (4)$$

P21 cont: I'll note $r(\phi)$ is identical to Kepler ellipse under transform

$$\theta = \beta(\phi - \phi_0)$$

(3.) Precession: Perhelian occurs when cos in $r(\phi) = 1$

↳ $\beta(\phi - \phi_0) = 2\pi n$; seperated by $\Delta\phi = \frac{2\pi}{\beta}$

for normal kepler: $\Delta\phi_K = 2\pi$ thus angular shift per orbit

$$\boxed{\delta\phi = 2\pi\left(\frac{1}{\beta} - 1\right)} \quad \dots (5)$$

for a weak Perturbation: $\left|\frac{2mh}{\ell^2}\right| << 1$

$$\beta^{-1} = \left(1 + \frac{2mh}{\ell^2}\right)^{-1/2} \approx 1 - \frac{mh}{\ell^2}$$

↳ $\boxed{\delta\phi \simeq -\frac{2\pi mh}{\ell^2}} \rightarrow |\Omega_{prec}| \approx \frac{2\pi m|h|}{\ell^2 T}$

(4.) mecury: $e = 0.206$, $T = 0.24$ yr : observed $\delta\phi_{100} = 40''$ per century

$\eta = \frac{h}{ka}$; $\ell^2 = mka(1-e^2)$

↳ $\frac{mh}{\ell^2} = \frac{mh}{mka(1-e^2)} = \frac{\eta}{1-e^2}$ || $N = \frac{100}{0.24} \approx 416.67$

$\delta\phi = \frac{40''}{416.67} \approx 0.096'' \cdot \left(\pi/64800\right)$ rad $= 4.65 \times 10^{-7}$ rad

$\eta \approx \frac{(1-e^2)\delta\phi}{2\pi} \approx -7.1 \times 10^{-8}$ ✓